

# Online Privacy Concerns and the Teenage Perspective: A Comprehensive Review and Next Steps Towards Teen-Centered Design of Social Media Platforms

This paper has been anonymized as part of the double-blind review process.

**Abstract.** Meeting teen users' privacy needs necessitates involving them design processes. Prior work reached this objective using smaller-scale qualitative methods (i.e. codesign). However, we believe quantitative methods have a role to play in the push towards teen-centered privacy design, enabling large-scale understanding of teenage privacy perceptions. This preliminary poster, therefore, focuses on the quantifiable concept of privacy concerns and reviews existing literature to discern a clear definition and methods of measuring it, understand how said methods vary across contexts, and assess the extent to which teenagers have been involved in said research. This study reveals the privacy concerns scale as an effective quantitative tool for understanding teenage users' evaluations while also revealing a major flaw with existing privacy concerns scales in that they have been primarily designed for adults, even those used to assess teenage populations, potentially failing to address the unique privacy concerns of teenagers.

**Keywords:** Privacy Concerns, Teenagers, Social Media.

## 1 Introduction

Research in information and computing systems shows a growing concern for a lack of incorporating users' perspectives when designing attributes that affect system's privacy [1, 2]. Despite this, it is still more common for developers to design systems' privacy based on consultation with other developers [3, 4], Chief Privacy Officers [3, 5], and based on pre-existing privacy policy statements [6–8] than it is for them to involve users in the design process. The result is a trend throughout privacy-related design processes where designers fail to consider users' perspectives and, consequently, leave users' privacy needs unsatisfied [9, 10].

This disconnect between users and designers is particularly illustrated by the misalignment between teenage users' privacy needs and the privacy features their adult counterparts design. Within the United States as of 2022, Pew Research Center reports that nearly all teens both have access to a smartphone and use the internet on a daily basis [11]. COVID has only increased this usage, with 72% of parents in the United States reporting a considerable increase in their K-12 kids' (this population including teenagers) screen time during and after the pandemic [12]. Despite these rates of access and daily use, there is a common misconception that teenagers do not care about their privacy online. Extant literature, however, suggests that teens actively

consider their online privacy, develop privacy management strategies, and are critical of privacy-related design features that don't meet their needs [11, 12, 13]. Teenagers – especially those with some degree of education surrounding privacy risk [14] – are highly attentive to the privacy features on websites they interact with. Yet as Wisniewski [14] points out, most of these features are designed with an adult-centered or parent-centered approach. This often leaves teens feeling considerably disappointed by privacy technologies that focus more on enabling parental control than they do on providing resources for teens to ensure their own safety [15]. Emphasizing adult control rather than teen resilience runs the risk of introducing a number of unintended consequences: denying teenagers the opportunity to autonomously learn how to ensure their own safety in online spaces, conceptualize boundaries and practice setting/maintaining them, and negotiate the balance between information sharing and protection of privacy common in online relationships, which are becoming increasingly prevalent [14]. The loss of these opportunities may have severe ramifications on teenagers, both online and offline, as they approach adulthood [18].

To alleviate these unintended negative consequences of excluding teens' perspectives and respect the agency teenagers possess in negotiating the privacy risks they may encounter online, Wisniewski argues for a shift away from adult or parent-centered privacy design to a teen-centered design approach. This approach prioritizes the teenage perspective and specifically asks teenagers what they feel they need to be kept safe online [14]. In other words, it is imperative to understand teenagers' privacy concerns in order to design systems that benefit and support them in developing and mastering the skills necessary to independently negotiate the risks that exist in both the virtual and physical spaces they occupy.

A considerable amount of work towards teen-centric privacy design has already been conducted and provides valuable insights into technological designs that could benefit teens in their efforts to autonomously protect their privacy [14, 18–21]. However, most of these existing studies have utilized smaller-scale, qualitative approaches towards investigating teenagers' privacy concerns which, while deeply valuable in terms of understanding teens' values and lived experiences, are limited in scale.

This poster is only the first phase in a larger, multi-phase research programme that ultimately asks, "How can we augment and support teen-centered, qualitative approaches to understanding teenage users' privacy concerns at scale? What are the opportunities to build on existing work and quantitatively elicit teen users' feedback to help social media platform designers understand these users' privacy concerns and ultimately better meet their privacy needs?" This research focuses on the context of teens' privacy concerns surrounding social media platforms specifically because Pew Research Center data from 2022 shows that the majority of teenagers in the U.S. are using a wide array of social media platforms 'almost constantly' [11].

While these are ambitious questions, this preliminary poster serves as the first phase and is a literature review broken into three parts. First is a review of how privacy concerns are defined and quantitatively measured across the existing literature. The second part explores how the measurement of privacy concerns changes between general measurement and measurement within the contexts of a specific platform, including and especially social media platforms. The third part reviews the extent to

and manners in which teens have been included in privacy concerns research up to this point. This Phase One literature review and the gaps identified shape the proposed next steps of the research programme discussed in the Future Work section.

## 2 Literature Review

### 2.1 An overview of privacy concerns constructs

As a construct, privacy concerns lacks a singular definition, as there have been multiple conceptualizations proposed across the literature [22–34]. Earlier conceptualizations argued a four-factor model composed of users’ (1) concerns for the amounts of information companies collect and the fact that they collect information to begin with (collection), (2) concerns for the measures companies take to protect collected information from unauthorized access, (3) concerns for the accuracy of the information stored by said companies (errors), and (4) concerns for the undisclosed ways in which companies may use or share the information (secondary use) [22, 23]. Other studies have conceptualized privacy concerns as a two-factor construct comprised of concern for how easily locatable submitted information is (privacy concerns for information finding) and concern for how said information could be abused by malicious entities (privacy concerns for information abuse) [24, 25]. A prominent conceptualization presented a conceptualization consisting of the aforementioned concern regarding collection, concern for how much control an individual has over their submitted information, and concern for the level of transparency offered by websites regarding the intended use of the information (awareness) [26].

Across the various conceptualizations exists a collection of prominent commonalities. The first of these is the near-unanimous agreement of privacy concerns as a multidimensional construct, the components of which should be measured with designated items on any privacy concerns scale going forward [22–27]. The second commonality amongst the proposed models is the repeated inclusion of awareness as a factor of privacy concerns. It is explicitly integrated in a number of highly relevant models [26, 27] and implicitly present in components of other models. Specifically, the concern for information privacy abuse [24, 25] and the concern for secondary use [23] both inherently entail a concern for how transparent websites are regarding their intended use of submitted information. Tangential studies further emphasize the importance of awareness, one noting how valued awareness is by those sharing information in healthcare systems [8] and the other covering how practices regarding the promotion of awareness by companies positively correlate to levels of perceived privacy risk and perceived privacy control [28]. Thirdly, concern for information collection is repeated as a factor of the construct in all but one of the conceptualizations [23, 26, 27], displaying the general agreement that collection is a definitive dimension of privacy concerns. The final repeated factor is the concern for control over information submitted. This concern is directly built into two of the four models [23, 26] and further highlighted by three other studies involving privacy concerns and privacy perceptions, which identify similar trends in which increased perceived priva-

cy control results in lowered privacy concerns and increased privacy safety [24, 28, 29]. However, the latter three studies raise the question of whether control is a component of privacy concerns or a correlate. Further reading will need to be done on this matter before definitively determining whether control is a reliable component of privacy concerns. All said, the literature ultimately makes clear the importance of conceptualizing privacy concerns as a multidimensional construct composed, at least, of the three factors of awareness, collection, and control.

## **2.2 General privacy concerns vs. Privacy concerns in context**

While there are a variety of scales in the extant literature that measure privacy attitudes and concerns generally, this review is scoped to studies that focus on measuring users' evaluation of privacy concerns within a specific context [1, 34–44]. Such studies used different privacy-related constructs, such as risk [1, 35], trust [35, 38], and system's privacy, as it is perceived by the users [1, 36, 38]. Some of the studies were focused on privacy scales in the context of social media [40–44], examining users' evaluation of the platforms' access [40], trust [42], control [22], risk [42], privacy concerns [43], [44] as well as protection strategies [41]. However, as we further elaborate in the next section, while useful as sources for constructs and questions, most of these studies focused on the adult population.

## **2.3 The exclusion of youth populations in scale development and validation**

Despite the extensive research exploring privacy concerns, including those studies investigating privacy concerns in youth populations [30, 31, 45], the constructs developed, and the measurement instruments (scales) derived from them, are overwhelmingly validated using adult populations. Such can be seen most prominently in the prevailing conceptualizations of privacy concerns highlighted in this review were all validated using sample populations of eighteen and older [23, 24, 26, 27]. Tangential studies investigating the role of privacy concerns in the wider privacy construct similarly used scales validated only with adult populations [28, 32, 36, 46, 47]. Even those studies which have assessed teenage privacy concerns and privacy perspectives in a quantitative manner [30, 31, 45] have all used scales (either directly imported or adapted from other studies) developed for and validated with adult populations [33, 48, 49]. In contrast, De Wolf et al. [49] included teenage participants, however, this work focuses on privacy management rather than privacy assessment, and the privacy concerns constructs used were still adapted from a scale created for adults [28].

Summatively, this collection of literature shows that across a wide plethora of studies on privacy concerns, either as a general concept [8, 23, 24, 26, 27, 30–32, 45–47] or within the context of specific systems [29, 34, 36–44, 50], there is a tendency to apply scales designed and validated with adult populations in the assessment of teenage populations. We highlight this misalignment in design because, despite the simi-

larities between adults and teenagers, as we reviewed in the Introduction, there are ultimately key differences between the two populations with regards to privacy. Since teens have different motivations to use social media [51], different privacy behaviors [52] and conceptions [53] compared to those of adults, there is a need to develop a dedicated scale, as previous scales might not cover all teen-relevant topics.

### 3 Future Work

This paper examines how privacy concerns are currently defined, measured, and applied to general and platform-specific contexts. While there is extensive literature in this area, the existing measurements and scales both generally and applied to a social media context focus on adult populations. Even scales that are applied to teenage populations are either directly exported from or adapted from scales originally designed for adult populations, potentially neglecting factors unique to teenage privacy concerns [30, 31, 45]. In many ways, this quantitative centering of adults' perspectives in measuring privacy concerns generally and within social media platforms mirrors the aforementioned critiques surrounding the exclusion of the teenage perspective in online platforms' privacy designs. However, means of measuring teenagers' online privacy perceptions are primarily adapted from scales developed for adults, potentially neglecting factors of their unique privacy concerns [33, 34, 48]. This reality puts teens at an increased risk for privacy breaches due to a lack of privacy measures provided for teens [33, 34, 48].

Considering the exclusion of teens' perspectives in existing quantitative measures of privacy concerns within a social media context, the direction of our future research will consist of two parts to address these gaps. First, we will systematically review qualitative studies that explore teenage users' online privacy concerns in ways that the current quantitative approaches reviewed in this paper do not, focusing on social media as the context. We will systematically collect and review these studies to identify both factors of privacy concerns that overlap with the adult-centered quantitative scales, as well as factors that are unique to teen populations. Second, we will utilize our findings to develop a teen-centered scale specifically designed to measure their privacy concerns and ultimately evaluate the privacy features of the social media platforms with which they interact. This scale will build on the strengths of existing qualitative work that centers teens' perspectives in ways the current adult-centered measurements do not, while enabling the collection of potentially thousands of users' perspectives in ways that are generalizable to large, diverse populations. This will ultimately enable designers to make thoroughly informed teen-centered decisions in their designs to bridge the current gap between teens' privacy needs and adult-centered privacy features on social media platforms.

### 4 Conclusion

Despite their unique set of privacy concerns and privacy management strategies, teenage users are often neglected in the privacy designs of online platforms, either due to

a lack of consideration from designers or privacy designs that prioritize parental control. While there is a growing call for a more teen-centered approach in online privacy design, most studies in this area have been smaller-scale, qualitative investigations. This preliminary poster investigates the role quantitative methods could play in the investigation of teenagers' privacy concerns, specifically in the context of social media platforms, and concludes: (1) the privacy concerns scale is a viable tool for understanding teenage privacy concerns at scale but (2) existing scales are either entirely imported or otherwise closely adapted from scales designed for adult users, potentially failing to properly address teenagers' unique privacy concerns. Such proves that there is work to be completed before we begin proper work on our overarching goal of developing a scale tailored towards the unique privacy concerns of teenagers. However, our hope is that these efforts will further progress towards teen-centered design on social media platforms and empower teenage users to autonomously negotiate the privacy complexities of their virtual spaces.

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